

SHRI G. S. INSTITUTE OF TECHNOLOGY AND
SCIENCE, INDORE



Department of Computer Technology and Applications
MCA First Year Semester-I
Session:- July 2026– December 2026

Subject:- C Programming

Assignment-2

Topic :- Decision Making

Submitted To:-
Prof. Khushi Solanki

Submitted By:-
Ritika Rathore

Questions on if

1. **Positive Number** - Write a C program to input a number and print "Positive" if the number is greater than 0.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    printf("Enter the number: ");
```

```
    scanf("%d", &n);
```

```
    if (n>0)
```

```
    {
```

```
        printf("Positive");
```

```
    }
```

```
    return 0;
```

```
}
```

Output:

```
Enter the number: 98
```

```
Positive
```

```
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

2. Even Number- Write a program to input a number and print "Even" if the number is divisible by 2

```
#include <stdio.h>

int main()
{
    int n;

    printf("Enter the number: ");
    scanf("%d", &n);

    if(n%2==0)
    {
        printf("Even");
    }

    return 0;
}
```

Output:

```
Enter the number: 234
Even
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

3. Divisible by 5- Write a program to input a number and print "Divisible by 5" if it is divisible by 5.

```
#include <stdio.h>

int main()
{
    int n;

    printf("Enter the number: ");
    scanf("%d", &n);

    if(n%5==0)
    {
        printf("Divisible by 5");
    }

    return 0;
}
```

Output:

```
Enter the number: 105
Divisible by 5
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

4. Positive and Even- Write a program to check whether a number is positive and even using if.

```
#include <stdio.h>

int main()
{
    int n;

    printf("Enter the number: ");
    scanf("%d", &n);

    if(n>0 && n%2==0)
    {
        printf("Positive and Even");
    }

    return 0;
}
```

Output:

```
Enter the number: 24
Positive and Even
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> |
```

5. Salary Check-Write a program to input salary and print "High Salary" if salary is greater than ₹50,000.

```
#include <stdio.h>

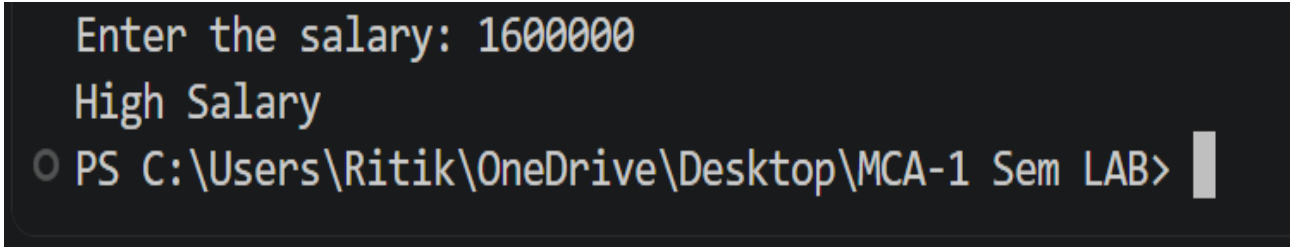
int main()
{
    int salary;

    printf("Enter the salary: ");
    scanf("%d", &salary);

    if(salary>50000)
    {
        printf("High Salary");
    }

    return 0;
}
```

Output:



```
Enter the salary: 1600000
High Salary
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

Questions on if-else

1. **Positive or Negative**- Write a program to input a number and check whether it is positive or negative.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    printf("Enter the number: ");
```

```
    scanf("%d", &n);
```

```
    if (n>0)
```

```
    {
```

```
        printf("Positive");
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("Negative");
```

```
    }
```

```
    return 0;
```

```
}
```

Output :

```
Enter the number: 46
```

```
Positive
```

```
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

Enter the number: -85

Negative

○ PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> █

2. Pass or Fail- Write a program to input marks and print:
Pass if marks are 40 *Or* above Fail otherwise.

```
#include <stdio.h>

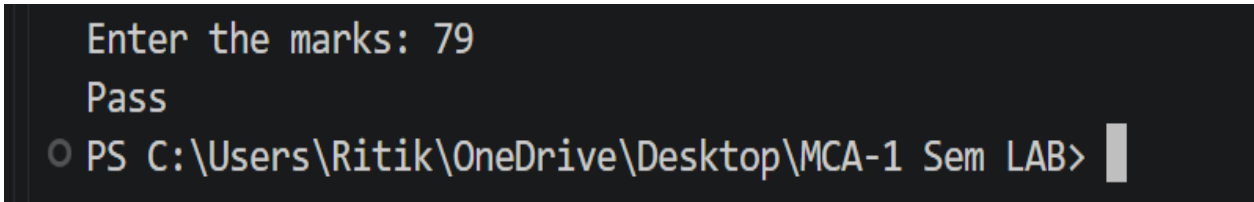
int main()
{
    int marks;

    printf("Enter the marks: ");
    scanf("%d", &marks);

    if(marks>=40)
    {
        printf("Pass");
    }
    else
    {
        printf("Fail");
    }

    return 0;
}
```

Output:



```
Enter the marks: 79
Pass
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

Enter the marks: 23

Fail

○ PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> █

3. Profit or Loss - Input cost price and selling price. Print whether there is a profit or loss.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int cp, sp;
```

```
    printf("Enter the cost price: ");
```

```
    scanf("%d", &cp);
```

```
    printf("Enter the selling price: ");
```

```
    scanf("%d", &sp);
```

```
    if(sp>cp)
```

```
    {
```

```
        printf("There is a profit");
```

```
    }
```

```
    else
```

```
    {
```

```
        printf("There is a loss");
```

```
    }
```

```
    return 0;
```

```
}
```

Output:

```
Enter the cost price: 230  
Enter the selling price: 250  
There is a profit  
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter the cost price: 180  
Enter the selling price: 150  
There is a loss  
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

4. Number is Zero or Not- Write a program to check whether a number is zero or non-zero.

```
#include <stdio.h>

int main()
{
    int n;

    printf("Enter the number: ");
    scanf("%d", &n);

    if(n==0)
    {
        printf("Number is zero");
    }
    else
    {
        printf("Number is non-zero");
    }

    return 0;
}
```

Output:

```
Enter the number: 0
Number is zero
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

Enter the number: 4

Number is non-zero

PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>

5. Divisible by 3- Write a program to check whether a number is divisible by 3 or not.

```
#include <stdio.h>

int main()
{
    int n;

    printf("Enter the number: ");
    scanf("%d", &n);

    if(n%3==0)
    {
        printf("Divisible by 3");
    }
    else
    {
        printf("Not divisible by 3");
    }

    return 0;
}
```

Output:

```
Enter the number: 987
Divisible by 3
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> |
```

Enter the number: 94

Not divisible by 3

○ PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB> █

Questions on else-if Ladder

1. **Grade Calculation**- Input marks and display:

90–100 → A

80–89 → B

70–79 → C

60–69 → D

Below 60 → F

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int marks;
```

```
    printf("Enter the marks: ");
```

```
    scanf("%d", &marks);
```

```
    if(marks>=90 && marks<=100)
```

```
    {
```

```
        printf("A");
```

```
    }
```

```
    else if(marks>=80 && marks<=89)
```

```
    {
```

```
        printf("B");
```

```
    }
```

```
else if(marks>=70 && marks<=79)
{
    printf("C");
}

else if(marks>=60 && marks<=69)
{
    printf("D");
}

else if(marks<60)
{
    printf("F");
}

else
{
    printf("Invalid Marks!!");
}

return 0;
}
```

Output :

```
Enter the marks: 99
```

```
A
```

```
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter the marks: 67
```

```
D
```

```
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

2. Number Range- Write a program to check whether a number is:

1–10 → Small

11–50 → Medium

51–100 → Large

Above 100 → Very Large

```
#include <stdio.h>
```

```
int main ()
```

```
{
```

```
    int n_range;
```

```
    printf("Enter the number: ");
```

```
    scanf("%d", &n_range);
```

```
    if(n_range>=1 && n_range<=10)
```

```
    {
```

```
        printf("Small");
```

```
    }
```

```
    else if(n_range>=11 && n_range<=50)
```

```
    {
```

```
        printf("Medium");
```

```
    }
```

```
    else if(n_range>=51 && n_range<=100)
```

```
    {
```

```
        printf("Large");
    }

    else if(n_range>100)
    {
        printf("Very Large");
    }

    else
    {
        printf("Invalid Number!!");
    }

    return 0;
}
```

Output:

```
Enter the number: 45
Medium
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

3. **Temperature**- Input temperature and display:

Above 40 → Very Hot

30–40 → Hot

20–29 → Normal

10–19 → Cold

Below 10 → Very Cold

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int temp;
```

```
    printf("Enter the temperature: ");
```

```
    scanf("%d", &temp);
```

```
    if (temp>40)
```

```
    {
```

```
        printf("Very Hot");
```

```
    }
```

```
    else if (temp>=30 && temp<=40)
```

```
    {
```

```
        printf("Hot");
```

```
    }
```

```
    else if (temp>=20 && temp<=29)
```

```
{  
    printf("Normal");  
}  
  
else if (temp>=10 && temp<=19)  
{  
    printf("Cold");  
}  
  
else  
{  
    printf("Very Cold");  
}  
  
return 0;  
}
```

Output:

```
Enter the temperature: 56  
Very Hot  
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter the temperature: 3  
Very Cold  
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

4. **Age Category**- Input age and display:

0–12 → Child

13–19 → Teenager

20–59 → Adult

60+ → Senior Citizen

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int age;
```

```
    printf("Enter the age: ");
```

```
    scanf("%d", &age);
```

```
    if (age>=0 && age<=12)
```

```
    {
```

```
        printf("Child");
```

```
    }
```

```
    else if (age>=13 && age<=19)
```

```
    {
```

```
        printf("Teenager");
```

```
    }
```

```
    else if (age>=20 && age<=59)
```

```
    {
```

```
        printf("Adult");
    }

    else if (age>=60)
    {
        printf("Senior Citizen");
    }

    else
    {
        printf("Invalid Age !!");
    }

    return 0;
}
```

Output:

```
Enter the age: 35
Adult
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter the age: 89
Senior Citizen
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

5. **Electricity Bill**- Input units and calculate the category:

0–100 → Low usage

101–200 → Medium usage

201–300 → High usage

Above 300 → Very high usage

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int units;
```

```
    printf("Enter the units: ");
```

```
    scanf("%d", &units);
```

```
    if(units >= 0 && units <= 100)
```

```
    {
```

```
        printf("Low usage");
```

```
    }
```

```
    else if(units >= 101 && units <= 200)
```

```
    {
```

```
        printf("Medium usage");
```

```
    }
```

```
    else if(units >= 201 && units <= 300)
```

```
    {
```

```
        printf("High usage");
```

```
}  
else if(units > 300)  
{  
    printf("Very high usage");  
}  
else  
{  
    printf("Invalid units");  
}  
  
return 0;  
}
```

Output:

```
Enter the units: 450  
Very high usage  
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter the units: -300  
Invalid units  
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

Questions on Nested if

1. **Positive Even/Odd-** Write a program using nested if to check whether a number is: Positive Even, Positive Odd, Negative.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    printf("Enter the number: ");
```

```
    scanf("%d", &n);
```

```
    if(n > 0)
```

```
    {
```

```
        if(n % 2 == 0)
```

```
        {
```

```
            printf("Positive Even");
```

```
        }
```

```
    else
```

```
    {
```

```
        printf("Positive Odd");
```

```
    }
```

```
}
```

```
else if(n < 0)
```

```
{
```

```
    printf("Negative");
```

```
}  
else  
{  
    printf("Zero");  
}  
  
return 0;  
}
```

Output:

```
Enter the number: 7  
Positive Odd  
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter the number: 96  
Positive Even  
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter the number: -87  
Negative  
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

2. Largest of Three Numbers- Write a program using nested if to find the largest among three numbers.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int a, b, c;
```

```
    printf("Enter three numbers: ");
```

```
    scanf("%d %d %d", &a, &b, &c);
```

```
    if(a > b)
```

```
    {
```

```
        if(a > c)
```

```
        {
```

```
            printf("%d is the largest", a);
```

```
        }
```

```
    else
```

```
    {
```

```
        printf("%d is the largest", c);
```

```
    }
```

```
}
```

```
else
```

```
{
```

```
    if(b > c)
```

```
    {
```

```
        printf("%d is the largest", b);
```

```
    }  
    else  
    {  
        printf("%d is the largest", c);  
    }  
}  
  
return 0;  
}
```

Output:

```
Enter three numbers: 3 4 5  
5 is the largest  
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

3. **Student Eligibility**- Input: Attendance percentage, Marks

A student is eligible for the exam only if attendance is at least 75%. If eligible, check whether the student has passed based on marks.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    float attendance, marks;
```

```
    printf("Enter attendance percentage: ");
```

```
    scanf("%f", &attendance);
```

```
    printf("Enter marks: ");
```

```
    scanf("%f", &marks);
```

```
    if(attendance >= 75)
```

```
    {
```

```
        if(marks >= 40)
```

```
        {
```

```
            printf("Eligible for exam and Passed");
```

```
        }
```

```
    else
```

```
    {
```

```
        printf("Eligible for exam but Failed");
    }
}
else
{
    printf("Not eligible for exam");
}

return 0;
}
```

Output:

```
Enter attendance percentage: 89
Enter marks: 47
Eligible for exam and Passed
○ PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter attendance percentage: 89
Enter marks: 34
Eligible for exam but Failed
○ PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter attendance percentage: 45
Enter marks: 99
Not eligible for exam
○ PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

Questions on switch-case

1. **Day of the Week**- Write a C program using switch-case to input number from 1 to 7 and display the corresponding day.

1 → Monday, 2 → Tuesday, 3 → Wednesday, 4 → Thursday, 5 → Friday, 6 → Saturday, 7 → Sunday

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int day;
```

```
    printf("Enter a number (1-7): ");
```

```
    scanf("%d", &day);
```

```
    switch(day)
```

```
    {
```

```
        case 1:
```

```
            printf("Monday");
```

```
            break;
```

```
        case 2:
```

```
            printf("Tuesday");
```

```
            break;
```

case 3:

printf("Wednesday");

break;

case 4:

printf("Thursday");

break;

case 5:

printf("Friday");

break;

case 6:

printf("Saturday");

break;

case 7:

printf("Sunday");

break;

default:

printf("Invalid input");

}

return 0;

}

Output:

```
Enter a number (1-7): 5
Friday
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter a number (1-7): 10
Invalid input
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

2. Vowel or Consonant- Write a program using switch-case to input a character and check whether it is a vowel (a, e, i, o, u) or not.

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char ch;
```

```
    printf("Enter a character: ");
```

```
    scanf(" %c", &ch);
```

```
    switch(ch)
```

```
    {
```

```
        case 'a':
```

```
        case 'e':
```

```
        case 'i':
```

```
        case 'o':
```

```
        case 'u':
```

```
        case 'A':
```

```
        case 'E':
```

```
        case 'I':
```

```
        case 'O':
```

```
        case 'U':
```

```
            printf("Vowel");
```

```
        break;

    default:
        printf("Not a vowel");
    }

    return 0;
}
```

Output:

```
Enter a character: A
Vowel
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter a character: a
Vowel
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter a character: z
Not a vowel
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

3. Number to Word-Write a program using switch-case to input a number from 0 to 9 and display it in words.Example: Input: 5, Output: Five

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    printf("Enter a number (0-9): ");
```

```
    scanf("%d", &n);
```

```
    switch(n)
```

```
    {
```

```
        case 0:
```

```
            printf("Zero");
```

```
            break;
```

```
        case 1:
```

```
            printf("One");
```

```
            break;
```

```
        case 2:
```

```
            printf("Two");
```

```
break;
```

```
case 3:
```

```
    printf("Three");
```

```
    break;
```

```
case 4:
```

```
    printf("Four");
```

```
    break;
```

```
case 5:
```

```
    printf("Five");
```

```
    break;
```

```
case 6:
```

```
    printf("Six");
```

```
    break;
```

```
case 7:
```

```
    printf("Seven");
```

```
    break;
```

```
case 8:
```

```
    printf("Eight");
```

```
    break;
```

```
    case 9:
        printf("Nine");
        break;

    default:
        printf("Invalid input");
}

return 0;
}
```

Output:

```
Enter a number (0-9): 7
Seven
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter a number (0-9): 10
Invalid input
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

4. Traffic Signal- Write a program using switch-case to input a character: R → Red → Stop, Y → Yellow → Wait, G → Green → Go

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char signal;
```

```
    printf("Enter the traffic signal (R/Y/G): ");
```

```
    scanf(" %c", &signal);
```

```
    switch(signal)
```

```
    {
```

```
        case 'R':
```

```
        case 'r':
```

```
            printf("Red - Stop");
```

```
            break;
```

```
        case 'Y':
```

```
        case 'y':
```

```
            printf("Yellow - Wait");
```

```
            break;
```

```
        case 'G':
```

```
    case 'g':  
        printf("Green - Go");  
        break;  
  
    default:  
        printf("Invalid signal");  
}  
  
return 0;  
}
```

Output:

```
Enter the traffic signal (R/Y/G): r  
Red - Stop  
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter the traffic signal (R/Y/G): Y  
Yellow - Wait  
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter the traffic signal (R/Y/G): g  
Green - Go  
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

5. Grade Description- Input a grade character (A, B, C, D, F) and display:

A → Excellent, B → Very Good, C → Good, D → Average,
F → Fail

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    char grade;
```

```
    printf("Enter the grade: ");
```

```
    scanf(" %c", &grade);
```

```
    switch(grade)
```

```
    {
```

```
        case 'A':
```

```
        case 'a':
```

```
            printf("Excellent");
```

```
            break;
```

```
        case 'B':
```

```
        case 'b':
```

```
            printf("Very Good");
```

```
            break;
```

```
case 'C':
```

```
case 'c':
```

```
    printf("Good");
```

```
    break;
```

```
case 'D':
```

```
case 'd':
```

```
    printf("Average");
```

```
    break;
```

```
case 'F':
```

```
case 'f':
```

```
    printf("Fail");
```

```
    break;
```

```
default:
```

```
    printf("Invalid grade");
```

```
}
```

```
return 0;
```

```
}
```

Output:

```
Enter the grade: C
Good
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter the grade: c
Good
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
Enter the grade: K
Invalid grade
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

6. Calculator with Nested Switch- Write a program using nested switch-case:

1. Arithmetic

1. Addition

2. Subtraction

3. Multiplication

4. Division

2. Relational

1. Greater

2. Smaller

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int choice, operation;
```

```
    float a, b;
```

```
    printf("1. Arithmetic\n");
```

```
    printf("2. Relational\n");
```

```
    printf("Enter your choice: ");
```

```
    scanf("%d", &choice);
```

```
    printf("Enter two numbers: ");
```

```
    scanf("%f %f", &a, &b);
```

```
    switch(choice)
```

```
    {
```

```
        case 1:
```

```
printf("1. Addition\n");
printf("2. Subtraction\n");
printf("3. Multiplication\n");
printf("4. Division\n");
printf("Enter your choice: ");
scanf("%d", &operation);

switch(operation)
{
    case 1:
        printf("Result = %.2f", a + b);
        break;

    case 2:
        printf("Result = %.2f", a - b);
        break;

    case 3:
        printf("Result = %.2f", a * b);
        break;

    case 4:
        if(b != 0)
            printf("Result = %.2f", a / b);
        else
            printf("Division by zero is not possible");
```

```
break;
```

```
default:
```

```
    printf("Invalid choice");
```

```
}
```

```
break;
```

```
case 2:
```

```
    printf("1. Greater\n");
```

```
    printf("2. Smaller\n");
```

```
    printf("Enter your choice: ");
```

```
    scanf("%d", &operation);
```

```
switch(operation)
```

```
{
```

```
    case 1:
```

```
        if(a > b)
```

```
            printf("%.2f is greater", a);
```

```
        else if(b > a)
```

```
            printf("%.2f is greater", b);
```

```
        else
```

```
            printf("Both numbers are equal");
```

```
        break;
```

```
    case 2:
```

```
        if(a < b)
```

```
        printf("%.2f is smaller", a);  
    else if(b < a)  
        printf("%.2f is smaller", b);  
    else  
        printf("Both numbers are equal");  
    break;
```

```
    default:  
        printf("Invalid choice");  
    }  
    break;
```

```
    default:  
        printf("Invalid choice");  
    }
```

```
    return 0;  
}
```

Output:

```
1. Arithmetic
2. Relational
Enter your choice: 1
Enter two numbers: 24 56
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice: 3
Result = 1344.00
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```

```
1. Arithmetic
2. Relational
Enter your choice: 2
Enter two numbers: 45 67
1. Greater
2. Smaller
Enter your choice: 1
67.00 is greater
PS C:\Users\Ritik\OneDrive\Desktop\MCA-1 Sem LAB>
```